Template Document

Academic Support Agent

February 19, 2025

Abstract

20 60 ER 90 Internet
.....

Small World, Network Model, Data Center
20 60 ER 90 Internet

1. WarpNet
   6 1000
2. DS SIDN
   SIDN
   /
3.
4.

20 60 ER 90 Internet

1. WarpNet
   6 1000
2. DS SIDN
   SIDN
   /
3.
4.

2013

1

Amazon EC2
7
PB
IO IaaS
IaaS
ESXi Xen KVM
ACL NAT
ARP

	Java	( )	
BOOLEAN	boolean	1	
CHAR	char	2	UTF-16
BYTE	byte	1	8
SHORT	short	2	16
INT	int	4	32
LONG	long	8	64
STRING	String		UTF-8
DATE	java.util.Date	8	GMT 1970 1 1 0 0 0
BYTE_ARRAY	byte[]		
BIG_INTEGER	java.math.BigInteger		
BIG_DECIMAL	java.math.BigDecimal		

Table 1

2

2.1

2.2

1 2 3 4 5 6 7 8 9 10

[**?**] [**?**, **?**, **?**] [**?**, **?**, **?**] [**?**, 124–128]

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  1.
  2.
  3.

1.
2.
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4. (a) i.
   ii.
   iii.

1
2
3
4
5
6
7
8
9
10

(b)

:

1

2

2.1

2.2

2.2.1

2.2.2

2.2.3

2.3

3

:

1

2

2.1

2.2

2.3

3

:

1

2

2.1

2.2

2.3

3

:

1

2

2.1

2.2

2.3

3

:

1

2

2.1

2.2

2.3

3

|quotel |quotel |quotel |quotel |quotel |quotel |quotel |quotel |quotel

|quotation| |quotation| |quotation| |quotation| |quotation| |quotation| |quotation| |quotation| |quotation|

[ ]

2.3

IaaS PaaS SaaS 70

1. Nvidia Georce GRID [**?**] GaaS[11] GPU

2.

web2.0

2.4

0

-0
1
-1
2
-2
3
-3
4
-4
5
-5
6
-6
7
8
0
-0
1
-1
2
-2
3
-3
4
-4
5
-5
6
-6
7
8
0
-0
1
-1
2
-2
3
-3
4
-4
5
-5
6
-6

[11] Gaming as a Service

7
8
0
-0
1
-1
2
-2
3
-3
4
-4
5
-5
6
-6
7
8
Huge
huge
LARGE
Large
large
normalsize
small
footnotesize
scriptsize
tiny
textsl
bfseries
textbf
textit
Fall-Out

$$\text{fallout} = \frac{|\{\ \} \cap \{\ \}|}{|\{\ \}|}$$

3

1 IaaS BR AR S

3.1 IaaS

3.2 PaaS

4

5

“ ” “ ”

1909 Guglielmo Marconi 5.83 6 [**?**] “ ” 6 [**?**]
1929 Frigyes Karinthy Everything is Different Chain-Links
[**?**, **?**] Karinthy “ ”
Karinthy “ ” Karinthy (six degrees of separation) [**?**]
1961 Michael Gurevich Ithiel de Sola Pool Manfred Kochen Sola Pool Contacts and Influences
[**?**] 1973 Gurevich Monte Carlo [**?**] Stanley Milgram

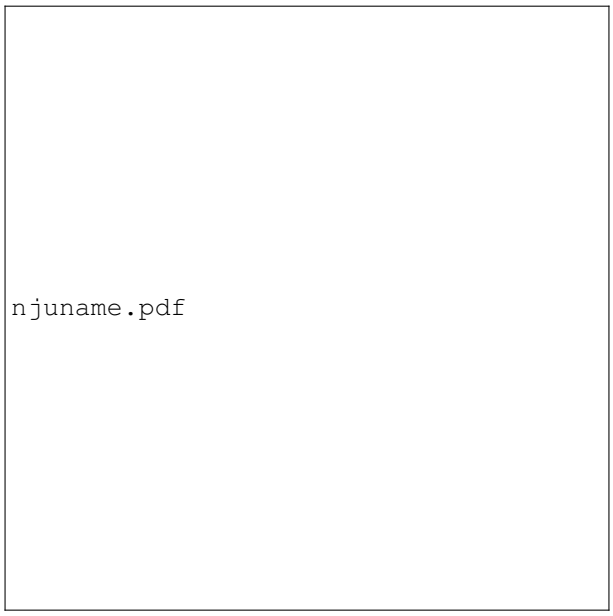


Figure 1

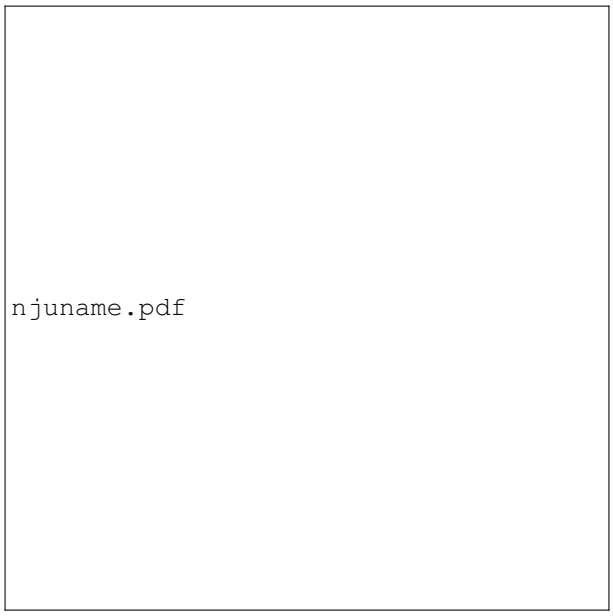


Figure 2

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Table 2

6

average path length clustering coefficient degree distribution Watts Strogatz

6.1

[ ] i j (distance)d_{ij}

[ ] D

$$D = \max_{i,j} d_{ij} \tag{1}$$

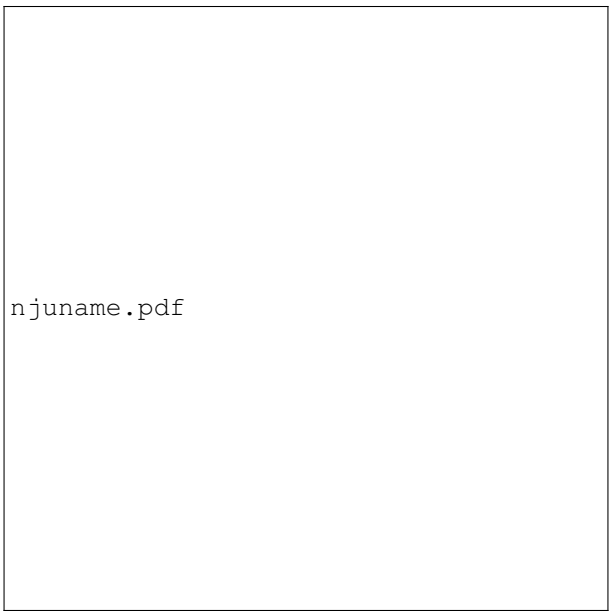


Figure 3

[ ] L

$$L = \frac{2}{N(N+1)} \sum_{i \geq j} d_{ij} \tag{2}$$

N

(2)

(2) $(N+1)/(N-1)$

6.2

(clustering coefficient)

[ ]

$$C_{total} = \frac{3 \times G_{\triangle}}{3 \times G_{\triangle} + G_{\wedge}}$$

C_{total} $G_{\triangle}$ $G_{\wedge}$ [**?**]

C_i Watts Strogatz 1998 [**?**]

[ ]

i k_i k_i i k_i $k_i(k_i-1)/2$ k_i E_i $k_i(k_i-1)/2$ i (clustering coefficient)C_i

$$C_i = \frac{2E_i}{k_i(k_i-1)} \tag{3}$$

$$C_i = \frac{i}{i} \tag{4}$$

i i

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Table 3

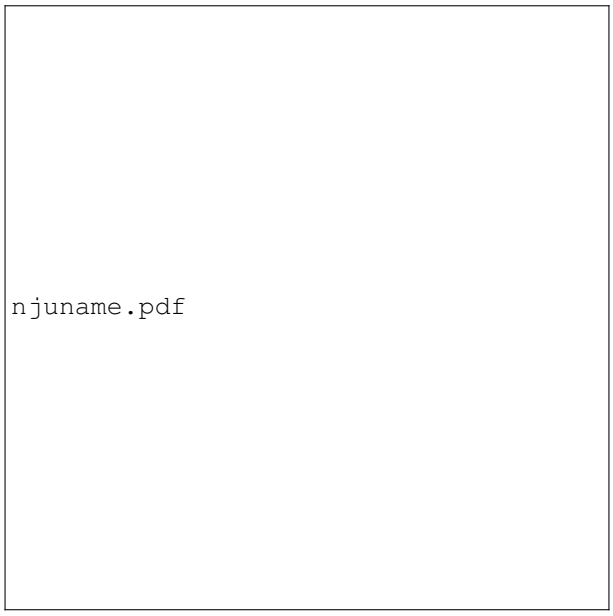


Figure 4

	Java	( )	
BOOLEAN	boolean	1	
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Table 4

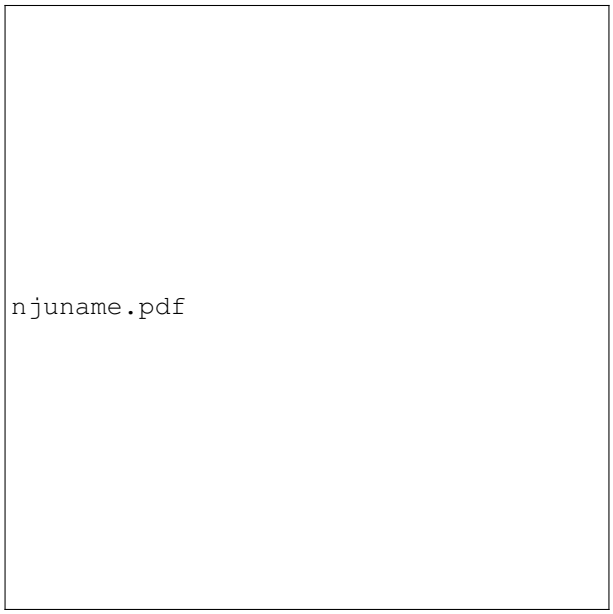


Figure 5

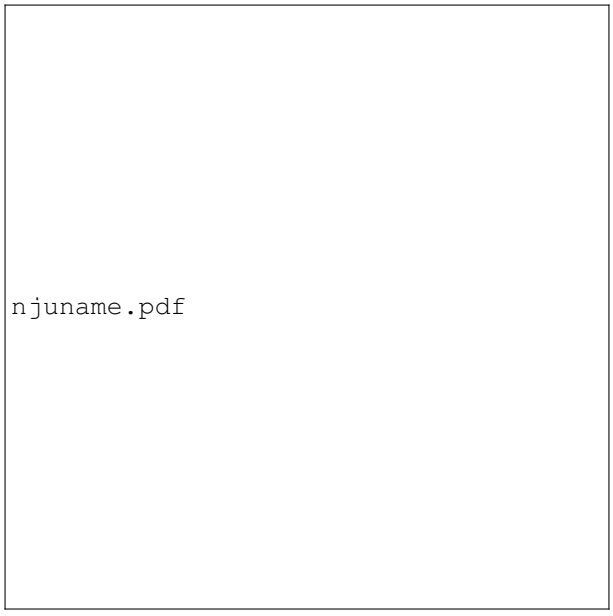


Figure 6

Watts Strogatz 1998 [**?**]
[ ]

$$\bar{C} = \frac{1}{n} \sum_{i=1}^{n} C_i. \tag{5}$$

6.3

[ ] i (degree)k_i (out-degree) (in-degree)
[ ] i k_i $Avgk$
[ ] $P(k)$

$$P(k) = \ k \tag{6}$$

$P(k)$

7

[ ]

$$L_G \propto \log N$$

N “ ”

8

8.1 DS

8.2 DS

8.3

9 SIDN

9.1 SIDN

9.2

10

10.1

10.2

11 SIDN

11.1

11.2

11.3

11.4

12

12.1

12.2

12.3

13

14

15 WarpNet

15.1

15.2

16

16.1 flood

17

17.1

17.2

17.3

17.4

18

19

	Java	( )	
BOOLEAN	boolean	1	
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BIG_DECIMAL	java.math.BigDecimal		

Table 5

SIDN
/
20 Scafida [**?**] BA
-
5
q
13
WarpNet
6 1000
28
( ) ( )

33

34

GB7713
GB7714

35

A4(210mm×297mm) 25mm 20mm

36

( ) ( )
( ) ( ) ( )
( )

37

37.1

1. UDC
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37.2

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10

37.3

37.4

1500 1500

37.5

3 8

37.6

37.7

38

38.1

38.2

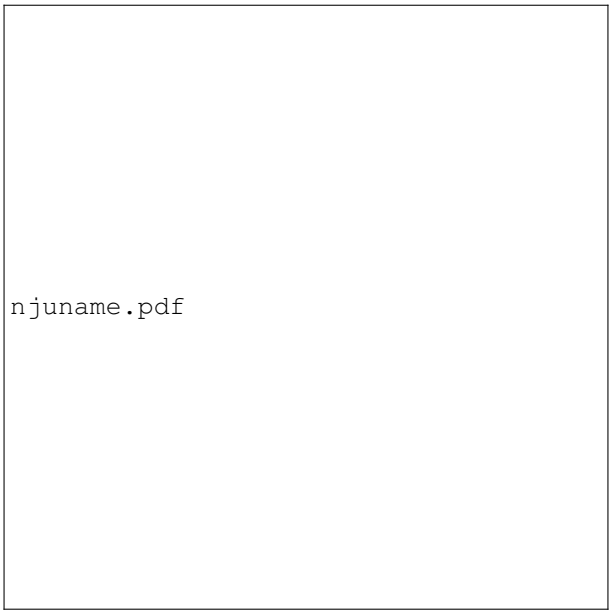


Figure 7

1.1 1.1.1
1.2 2.3 1 [4] (6.3)

1

$$C_i = \frac{2E_i}{k_i(k_i - 1)} \tag{7}$$

A B C $\cdots$ A
A.1 B.2 (B.3) [A.5]

38.3

$1 \sim 2$ 1 100 60%

38.4



Figure 8

( )

38.4.1

( )

38.4.2

xxx[1)] *
$\ldots 0$

38.4.3

( )
$+ - \times \div < > =$
. 999 0
94 652.023 567 0.314 325, $94,652.023,567$ $.314,325$

38.4.4

B

38.4.5

38.5

38.6

-
-
-
-
-

38.7

38.7.1

1. 2 1962

38.7.2

2. 1973 (4) 400 429

39

1.
2.
3.
4.
5.

40 ( )

1985 11

Xiaobao Wei, Jinnan Chen, “Voting-on-Grid Clustering for Secure Localization in Wireless Sensor Networks,” in *Proc. IEEE International Conference on Communications (ICC) 2010*, May. 2010.

Xiaobao Wei, Shiba Mao, Jinnan Chen, “Protecting Source Location Privacy in Wireless Sensor Networks with Data Aggregation,” in *Proc. 6th International Conference on Ubiquitous Intelligence and Computing (UIC) 2009*, Oct. 2009.

“ ” 2010 1 — 2012 12

“ ” 2010 1 — 2010 12 LTE/SAE